

Name: Chiranjeeb Behera

Regn No: 2141002044

DAY1

```
chira@LAPTOP-I6PG7VUO:~/hjk$ g++ day1.cpp -o day1
chira@LAPTOP-I6PG7VUO:~/hjk$ ./day1
Total RAM: 7807MB
Total RAM: 7GB
Free RAM: 6884MB
Free RAM: 7GB
chira@LAPTOP-I6PG7VUO:~/hjk$ |
```

C: > Users > chira > Desktop > g > day1 > new.cpp

```
1  #include<iostream>
2  #include<sys/sysinfo.h>
3  using namespace std;
4
5  void displayMemoryInfo(){
6      struct sysinfo info;
7      if(sysinfo(&info)==0){
8          cout<<"Total RAM: "<<info.totalram/(1024*1024)<<"MB\n";
9          cout<<"Total RAM: "<<info.totalram/(1024*1024*1024)<<"GB\n";
10         cout<<"Free RAM: "<<info.freeram/(1024*1024)<<"MB\n";
11         cout<<"Free RAM: "<<info.freeram/(1024*1024*1024)<<"GB\n";
12     }
13 }
14
15 int main(){
16
17     displayMemoryInfo();
18
19     return 0;
20 }
21
```

DAY2

```
chira@LAPTOP-I6PG7VU0:~/hjk/day2$ g++ day2.cpp -o day2
chira@LAPTOP-I6PG7VU0:~/hjk/day2$ ./day2
CPU USAGE:89.6904%
user Time:6743
Total cpu Time:1287409
cpu usage:0
chira@LAPTOP-I6PG7VU0:~/hjk/day2$ |
```

```
new.cpp x
C: > Users > chira > Desktop > g > day2 > new.cpp
1  #include<iostream>
2  #include<fstream>
3  #include<sstream>
4  #include<thread>
5  #include<chrono>
6  using namespace std;
7
8  struct CPUData
9  {
10 long user,nice,system,idle,iowait,irq,softirq,steal,guest,guest_nice;
11 };
12
13 CPUData getCPUData(){
14
15 ifstream file("/proc/stat");
16
17 string line;
18 CPUData cpu={};
19
20 if(file.is_open()){
21 getline(file,line); //read first line
22 istringstream ss(line);
23
24 string cpuLabel;
```

```
actpocc.cpp X
C: > Users > chira > Desktop > g > day2 > actpocc.cpp
1  #include <iostream>
2  #include <filesystem>
3  #include <fstream>
4  #include <sstream>
5  #include <algorithm>
6
7
8  namespace fs = std::filesystem;
9
10 bool isNumber(const std::string &s) {
11     return !s.empty() && std::all_of(s.begin(), s.end(), ::isdigit);
12 }
13
14 std::string getProcessName(int pid) {
15     std::ifstream file("/proc/" + std::to_string(pid) + "/stat");
16     std::string line, processName;
17     if (file.is_open()) {
18         std::getline(file, line);
19         std::istringstream ss(line);
20         std::string token;
21         int count = 0;
22         while (ss >> token) {
23             count++;
24             if (count == 2) {
```

DAY3

```
chira@LAPTOP-I6PG7VU0:~/hjk/day3$ g++ day3.cpp -o day3
chira@LAPTOP-I6PG7VU0:~/hjk/day3$ ./day3
PID      CPU%      Memory (kb)   Name
1         0.0655117%    4607          (systemd)
93        0.023327%    4607          (systemd-udevd)
51        0.0204111%    4607          (systemd-journal)
103       0.0160373%    11988         (systemd-resolve)
104       0.0145794%    6644          (systemd-timesyn)
179       0.0145794%    6644          (wsl-pro-service)
208       0.0131214%    6644          (unattended-upgr)
349       0.0116805%    11476         (systemd)
176       0.0116635%    6644          (systemd-logind)
297       0.0102205%    5412          (bash)
chira@LAPTOP-I6PG7VU0:~/hjk/day3$
```

```
day3.cpp X
C: > Users > chira > Desktop > g > day3 > day3.cpp
1 // display cpu and memory usage per process.
2 // sort porcesses by cpu and memory uses.
3 // total cpu time = utime + stime
4
5 // cpu usage+(utime+stime)/system uptime - starttime*100
6
7 #include <iostream>
8 #include <fstream>
9 #include <sstream>
10 #include <vector>
11 #include <algorithm>
12 #include <unistd.h> // for getpid() etc
13 #include <dirent.h> // For directory handling
14 #include <cstring> // For strcmp
15 #include <csignal> //for kill() and sigkill
16
17 struct Processinfo {
18     int pid;
19     std::string name;
20     double cpuUsage;
21     long memoryUsage;
22 };
23
24 std::string readFileValue(const std::string &path) {
```

DAY4

```
new.cpp X
C: > Users > chira > Desktop > g > day4 > new.cpp > ...
1 #include <iostream>
2 #include <fstream>
3 #include <sstream>
4 #include <vector>
5 #include <algorithm>
6 #include <unistd.h> // for getpid() etc
7 #include <dirent.h> // For directory handling
8 #include <cstring> // For strcmp
9 #include <csignal> //for kill() and sigkill
10
11 struct Processinfo {
12     int pid;
13     std::string name;
14     double cpuUsage;
15     long memoryUsage;
16 };
17
18 std::string readFileValue(const std::string &path) {
19     std::ifstream file(path);
20     std::string value;
21     if (file.is_open()) {
22         std::getline(file, value);
23     }
24     return value;
```

```

chira@LAPTOP-I6PG7VU0:~/hjk/day4$ ./day4
PID      CPU%      Memory (kb)   Name
1         0.0490866%    4607          (systemd)
93        0.0192283%    4607          (systemd-udev)
51        0.0160236%    4607          (systemd-journal)
296       0.0128326%    6644          (Relay(297))
103       0.0117506%    11988         (systemd-resolve)
104       0.0106824%    6644          (systemd-timesyn)
179       0.0106824%    6644          (wsl-pro-service)
297       0.00962443%   5412          (bash)
208       0.00961415%   6644          (unattended-upgr)
349       0.00855505%   11476         (systemd)
Enter PID to kill :51
Failed to kill process: Operation not permitted
chira@LAPTOP-I6PG7VU0:~/hjk/day4$ |

```

DAY5

```

chira@LAPTOP-I6PG7VU0:~/hjk/day4$ ./day4
PID      CPU%      Memory (kb)   Name
453       0.0346074%    5312          (Relay(454))
1         0.0338014%    4607          (systemd)
93        0.0162079%    4607          (systemd-udev)
454       0.0153811%    5180          (bash)
51        0.0112751%    4607          (systemd-journal)
179       0.0084563%    6644          (wsl-pro-service)
103       0.00775161%   11988         (systemd-resolve)
104       0.00704692%   6644          (systemd-timesyn)
208       0.00634223%   6644          (unattended-upgr)
349       0.00564151%   11476         (systemd)
Enter PID to kill :|

```

new.cpp

C: > Users > chira > Desktop > g > day5 > new.cpp

```
1  #include <iostream>
2  #include <fstream>
3  #include <sstream>
4  #include <vector>
5  #include <algorithm>
6  #include <unistd.h> // for getpid() etc
7  #include <dirent.h> // For directory handling
8  #include <cstring> // For strcmp
9  #include <csignal> //for kill() and sigkill
10 #include <thread>
11 #include <chrono>
12
13 struct Processinfo {
14     int pid;
15     std::string name;
16     double cpuUsage;
17     long memoryUsage;
18 };
19
20 std::string readFileValue(const std::string &path) {
21     std::ifstream file(path);
22     std::string value;
23     if (file.is_open()) {
24         std::getline(file, value);
```